Patient Retention Strategy : Churn Analysis and Prediction

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Churn Definition:

- Our basic time based churn definition was having no completed appointments within one year of the reference date, or 365 days to be more specific. We believe this is the best way to define churn on a time basis because each patient should at the very least be getting an annual checkup with the clinic.We considered shorter time frames, but we were concerned that we may define people who are in good health and only getting annual checkups as churners.

- This basic definition is good, but it lacks the ability to address some of the nuances in the enrollment data. For example, some people stop using the clinic because they started to qualify for Medicare or Medicaid. We argue that it is inaccurate to define these patients as churned because they simply fell outside of [redacted]'s area of coverage instead of making the conscious choice to stop using the clinic. To address this, we added a few extra criteria to our churn definition.

1. *[Simple Definition]* Any patient that formally dis-enrolls is not defined as churn unless they were dis-enrolled for compliance (as in they were forcibly dis-enrolled due to repeated missed appointments). The other formal disenrolment reasons, such as Medicare, Medicaid, and private insurance were not included in the churn definition.

2. *[Complex Definition]* Patients who let their enrollment expire instead of formally disenrolling were considered as churned if they did not have an expiration reason or if they expired due to lack of interest. This excludes expiration reasons such as Medicare and Medicaid like before. For anyone not disenrolled or expired, they are considered churned if they have not had a completed encounter in more than a year based on the reference date.

- For the purely time-based definition (simple): 4,672 out of the 6,172 patients (the count after removing dental and repeats) in our records were considered as churned.

- For the definition including disenrolment and expiration considerations (2) - 3,125 of the 6,172 were considered as churned.

- This means that approximately 75.7% of patients are considered as churned under the simple definition and 50.6% of patients are considered as churned under the complex definition.

Summary

- The features for our refined model were average number of visits per days enrolled.
- Counts of diagnoses by severity (5 features), and proportion of encounters that are no-shows.
- We ran several additional models including demographic data (I.E. Race, Language, Employment, etc.), but they did not generate consistent improvement that was worth the increase in complexity.
- This model had an accuracy of 66.3%, a sensitivity of 81.3%, and a specificity of 53.1%. Sensitivity is the most important
- Metric because the cost of false negatives is high whereas the cost of false positives is much lower.
- Furthermore, because our definition of churn is backwards-looking, there is always the possibility that a patient has demonstrated behaviors consistent with other churners but simply has not had a long enough lapse in appointments to be considered churned. This is especially pertinent for patients who have enrolled within the past 12 months of the reference date #as it is impossible for them to be considered as churned as they could not have expired or had a lapse in appointments.

Setup

```
rm(list=ls())

library(dplyr)
library(fpp3)
library(tidyverse)
library(lubridate)
```

Data Importing & Cleaning

- Our model focuses on 4 data sets: People, Enrollments, Encounters and Diagnosis

- What we did to the data
  - Remove Dental-only patients as well as patients who had closed or denied applications from Enrollment, People and Diagnosis
  - Remove duplicate IDs from people
  - Joining the people and enrollment data
  - Convert the dates to a compatible format
  - Finding the last encounter date for the patients

```
id = read_csv("people_updated.csv", )
enroll = read_csv('enrollment.csv')
encounter = read_csv('encounters.csv')
diagnosis <- read.csv("diagnosis.csv", stringsAsFactors = FALSE
  select(mrn_pseudo, dia_dxgroup)

enroll = enroll %>% filter(enr_service != 2)
unique_nondental = enroll %>% distinct(mrn_pseudo)
id = id%>% filter(mrn_pseudo %in% unique_nondental$mrn_pseudo)
id = id %>% distinct(mrn_pseudo, .keep_all = TRUE)
id_enroll = left_join(id, enroll, by = 'mrn_pseudo')
encounter = encounter %>% mutate(enc_date = mdy(enc_date))
```

```
Warning: There was 1 warning in `mutate()`.
ℹ In argument: `enc_date = mdy(enc_date)`.
Caused by warning:
! All formats failed to parse. No formats found.
```

```
enc_id_test = encounter %>% distinct(mrn_pseudo)
encounter = encounter %>% mutate(comp_date = case_when(enc_statu
encounter <- encounter %>%
  group_by(mrn_pseudo) %>%
  mutate(
    last_enc =
      if (all(is.na(comp_date))) {
        NA
      } else {
        max(comp_date, na.rm = TRUE)
      }
  ) %>%
  ungroup()
encounter = encounter %>% group_by(mrn_pseudo)%>% mutate(n_encou
enc_churn = encounter %>% select(mrn_pseudo, n_encounter, last_e
enc_churn = enc_churn %>% distinct(mrn_pseudo, .keep_all = TRUE

ref_date = ymd('2025-08-20')
churn_set = left_join(id_enroll, enc_churn, by = 'mrn_pseudo')
```

```
churn_set = churn_set %>%
  mutate(last_enc = case_when(is.na(last_enc)~ enr_start, TRUE ~
churn_set = churn_set %>%
  filter(!grepl('Closed', enr_status))%>%
  filter(!grepl('Denied', enr_status ))
```

Implementing Churn Definition

- The following code assigns a churn status to each patient for the complex and simple definitions of churn
- We also prep the no-show metric and join it to the churn-set here

```
churn_set = churn_set %>%
  mutate(has_churn = case_when(enr_status == 'Disenrolled' &
                                 enr_unenr_reason == 'Compliance
                               enr_status == 'Expired' |
                                 enr_status == 'Expired - No lon
                               as.numeric(ref_date-last_enc) > 3
                                 enr_status == 'Active' ~'Yes',

churn_set$has_churn <- factor(churn_set$has_churn, levels = c('N

#establish simple churn definition with just time consideration
churn_set = churn_set %>%
  mutate(has_churn_simple = case_when(as.numeric(ref_date-last_e
```

Proportion of no-shows (feature engineering)

- We use proportion of no-shows as one of our features because if a patient is frequently missing appointments, it is plausible that they would stop making appointments altogether.

```
#taking the number of no shows for each patient divided by their
no_show = encounter %>%
  select(mrn_pseudo, enc_status, enc_canc_reason)%>%
  group_by(mrn_pseudo)%>%
  mutate(n_encounters = n())%>%
  mutate(enc_status = case_when(enc_canc_reason == 'No Show/Late
  mutate(n_no_show = sum(enc_status == "No Show") )%>%
  mutate(n_no_show = case_when(is.na(n_no_show) ~ 0, TRUE ~ n_no
  mutate(prop_no_show = n_no_show/n_encounters)%>%
```

```
  select(mrn_pseudo, n_no_show, prop_no_show)%>%
  distinct(mrn_pseudo, .keep_all = TRUE)

#join the no-show data to the main table
churn_set = left_join(churn_set, no_show, by = 'mrn_pseudo')
```

Classifying the diagnosis metrics (feature engineering)

- Count the number of diagnoses by severity for each patient.
- This was originally just maximum severity but we felt that counting would give us more granularity to distinguish between patients with frequent low severity encounters and just one or two high severity and one with multiple high severity.
- This also will give us a better visibility on the effect of individual severity levels on chance of churn.

```
#classifications of diagnoses into more condensed categories, p
diagnosis <- diagnosis %>%
  mutate(simplified = case_when(
    dia_dxgroup == "Other" ~ "Other",
    dia_dxgroup == "Musculoskeletal" ~ "Musculoskeletal",
    dia_dxgroup == "Tobacco Use" ~ "Mental",
    dia_dxgroup == "Depression / Anxiety" ~ "Mental",
    dia_dxgroup == "Endocrine" ~ "Endocrine",
    dia_dxgroup == "Substance Use" ~ "Mental",
    dia_dxgroup == "Dyslipidemia" ~ "Cardiovascular",
    dia_dxgroup == "Other Circul Diseases" ~ "Cardiovascular",
    dia_dxgroup == "Other Behavioral Health" ~ "Mental",
    dia_dxgroup == "Chronic Pain" ~ "Chronic",
    dia_dxgroup == "Hypertension" ~ "Cardiovascular",
    dia_dxgroup == "Obesity" ~ "Obesity",
    dia_dxgroup == "Diabetes" ~ "Diabetes",
    dia_dxgroup == "Pre-diabetes" ~ "Diabetes",
    dia_dxgroup == "Asthma/COPD" ~ "Pulmonary",
    dia_dxgroup == "Heart Disease" ~ "Cardiovascular",
    dia_dxgroup == "Cerebrovascular Disease" ~ "Cerebrovascular
    TRUE ~ "Other"))

#estimated severities for each diagnosis, other is treated as a
diagnosis <- diagnosis %>%
  mutate(severity = case_when(
    dia_dxgroup == "Other" & simplified == "Other" ~ 1,
    dia_dxgroup == "Musculoskeletal" & simplified == "Musculosk
    dia_dxgroup == "Tobacco Use" & simplified == "Mental" ~ 1,
```

```
    dia_dxgroup == "Depression / Anxiety" & simplified == "Menta
    dia_dxgroup == "Endocrine" & simplified == "Endocrine" ~ 3,
    dia_dxgroup == "Substance Use" & simplified == "Mental" ~ 3,
    dia_dxgroup == "Dyslipidemia" & simplified == "Cardiovascula
    dia_dxgroup == "Other Circul Diseases" & simplified == "Car
    dia_dxgroup == "Other Behavioral Health" & simplified == "Me
    dia_dxgroup == "Chronic Pain" & simplified == "Chronic" ~ 2,
    dia_dxgroup == "Hypertension" & simplified == "Cardiovascula
    dia_dxgroup == "Obesity" & simplified == "Obesity" ~ 1,
    dia_dxgroup == "Diabetes" & simplified == "Diabetes" ~ 3,
    dia_dxgroup == "Pre-diabetes" & simplified == "Diabetes" ~ 2
    dia_dxgroup == "Asthma/COPD" & simplified == "Pulmonary" ~ 2
    dia_dxgroup == "Heart Disease" & simplified == "Cardiovascu
    dia_dxgroup == "Cerebrovascular Disease" & simplified == "Ce
    TRUE ~ 0))

diagnosis_count <- diagnosis %>%
  group_by(mrn_pseudo) %>%
  summarize(count_1 = sum(severity == 1), count_2 = sum(severity
  ungroup()

#join severity data to main data
churn_set = churn_set %>%
  left_join(diagnosis_count, by = "mrn_pseudo") %>%
  mutate(count_1 = ifelse(is.na(count_1), 0, count_1))%>%
  mutate(count_2 = ifelse(is.na(count_2), 0, count_2))%>%
  mutate(count_3 = ifelse(is.na(count_3), 0, count_3))%>%
  mutate(count_4 = ifelse(is.na(count_4), 0, count_4))%>%
  mutate(count_5 = ifelse(is.na(count_5), 0, count_5))

#additional cleaning for unused features
churn_set$Employment = case_when(grepl('Unemployed', churn_set$E
```

Education (feature engineering)

education was included as a demographic feature so that not all features relied on behavioral data and as a test to see if demographics could be explanatory of churn rates in theory patients with different levels of education will have different occupations and life styles which could create differences in churn rates.

```
churn_set = churn_set %>%
  mutate(Education = case_when(grepl('college', Education)~ 'Co
                               grepl('College', Education)~'Col
  mutate(Education = factor(Education))
```

Average lifetime clinic visits per year (feature engineering)

This is included as the most direct way to measure patient engagement. A patient who engages frequently is in theory more likely to keep engaging. Takes the average encounters per day enrolled based on start date to expiration date or start date to ref date. NA values can be replaced with zero *impossible and outlier values removed* Average of 12.16 visit per day, median of 5.38 visits per day

```
churn_set = churn_set %>%
  mutate(average_visits = case_when(enr_status == 'Active'~ n_e
churn_set = churn_set %>%
  mutate(average_visits = case_when(is.na(average_visits)~ 0 ,
churn_set = churn_set %>%
  filter(average_visits <= 150 & average_visits >= 0)

churn_set %>%
  summarize(average = mean(average_visits, na.rm = TRUE), media
```

```
# A tibble: 1 × 2
  average median
    <dbl>  <dbl>
1    12.6   6.02
```

Count of Severity_1 diagnoses

- Average of 20.26, median of 0

```
churn_set %>%
  summarize(average = mean(count_1, na.rm = TRUE), median = med
```

```
# A tibble: 1 × 2
  average median
    <dbl>  <dbl>
1    20.3      0
```

Count of Severity_2 diagnoses

- Mean of 3.21, median of 0

```
churn_set %>%
  summarize(average = mean(count_2, na.rm = TRUE), median = med
```

```
# A tibble: 1 × 2
  average median
    <dbl>  <dbl>
1    3.22      0
```

Count of Severity_3 diagnoses

- average of 3.76, median of 0

```
churn_set %>%
  summarize(average = mean(count_3, na.rm = TRUE), median = med
```

```
# A tibble: 1 × 2
  average median
    <dbl>  <dbl>
1    3.77      0
```

Count of Severity_4 diagnoses

- Mean of 2.63, median of 0

```
churn_set %>%
  summarize(average = mean(count_4, na.rm = TRUE), median = med
```

```
# A tibble: 1 × 2
  average median
    <dbl>  <dbl>
1    2.64      0
```

Count of Severity_5 diagnoses

- Mean of .056, median of 0

```
churn_set %>%
  summarize(average = mean(count_5, na.rm = TRUE), median = med
```

```
# A tibble: 1 × 2
  average median
    <dbl>  <dbl>
1  0.0537      0
```

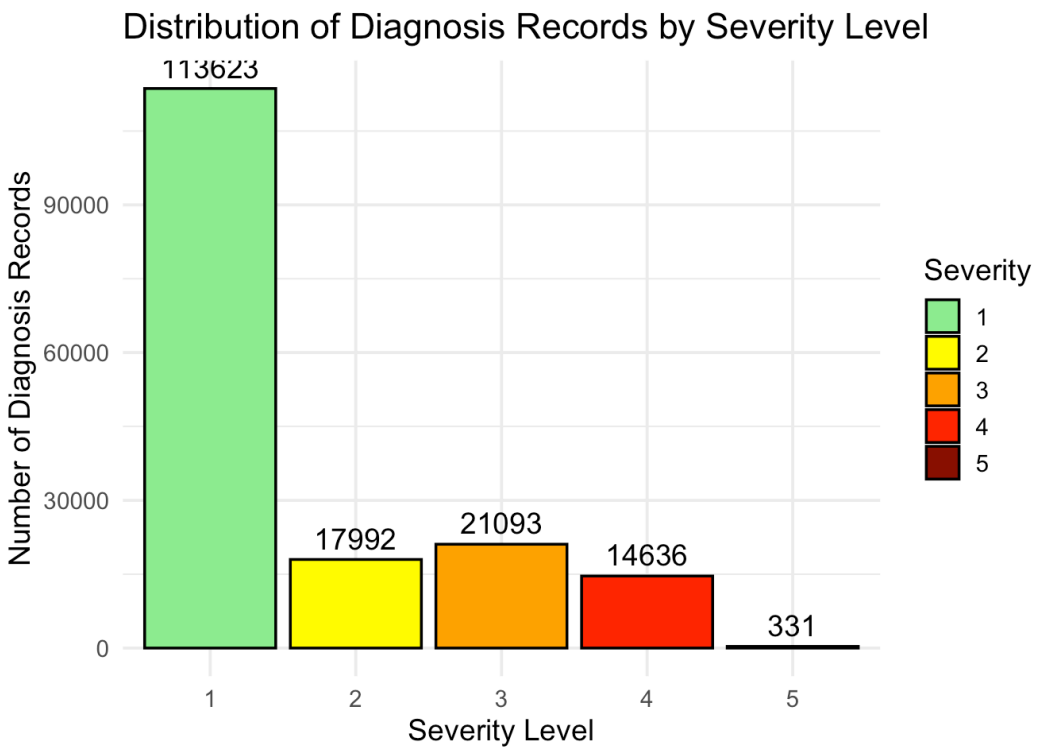
Visualization : Diagnosis distribution by

severity

- Severity 1 : Light-rashes, Dry eye
- Severity 2 : Chronic conditions / Hypertension /Pre-Diabetes
- Severity 3 : Diabetes / Endocrine
- Severity 4 : Musculoskeletal / Cardiovascular
- Severity 5 : Cerebrovascular

```
diagnosis %>% group_by(severity) %>%
  summarize(Count = n()) %>%
  mutate(Severity = as.character(severity)) %>%  ggplot(aes(x =
  geom_bar(stat = "identity", color = "black", size = 0.8) +
  scale_fill_manual(values = c("1" = "lightgreen",
                               "2" = "yellow",
                               "3" = "orange",
                               "4" = "red",
                               "5" = "darkred")) +
  geom_text(aes(label = Count), vjust = -0.5, linewidth = 4) +
  labs(
    title = "Distribution of Diagnosis Records by Severity Leve
    x = "Severity Level",
    y = "Number of Diagnosis Records") +
  theme_minimal(base_size = 14)
```

```
Warning in geom_bar(stat = "identity", color = "black", size =
0.8): Ignoring
unknown parameters: `size`

Warning in geom_text(aes(label = Count), vjust = -0.5,
linewidth = 4): Ignoring
unknown parameters: `linewidth`
```



Proportion of no-show encounters

- Average of .0048, median of 0

```
churn_set = churn_set %>%
  mutate(prop_no_show = case_when(is.na(prop_no_show)~ 0, TRUE

churn_set %>%
  summarize(average = mean(prop_no_show, na.rm = TRUE), median
```

```
# A tibble: 1 × 2
  average median
    <dbl>  <dbl>
1 0.00517      0
```

#Replace NA values for education with unknown if necessary *perform similar cleaning process for other demographics that were used in further testing

```
churn_set = churn_set %>%
  mutate(Education = case_when(is.na(Education) ~ 'Unknown', TRU

churn_set$Employment = case_when(is.na(churn_set$Employment)~'Un
```

```
churn_set$Employment = case_when(grepl('Choosing', churn_set$Em

churn_set$Gender = case_when(is.na(churn_set$Gender) ~ 'Unknown
```

Quality Check before creating and executing the Model

```
quality_check <- data.frame(
  Check = c("Entries where enrollment start/init is not availab
            "Number of duplicate rows",
            "Number of duplicate mrn ids",
            "Number of duplicate encounters",
            "Check for incorrect average visits",
            "Check for the Proportion of no_shows where value i
  Count = c(nrow(churn_set %>% filter(is.na(enr_start) & is.na(
            sum(duplicated(churn_set)),
            nrow(churn_set %>% group_by(mrn_pseudo) %>% filter(
            nrow(enc_churn %>% group_by(mrn_pseudo) %>% filter(
            sum(churn_set$average_visits < 0, na.rm = TRUE),
            sum(churn_set$prop_no_show > 1 | churn_set$prop_no_

print(quality_check, right = FALSE, row.names = FALSE)
```

```
 Check
Count
 Entries where enrollment start/init is not available         0
 Number of duplicate rows                                     0
 Number of duplicate mrn ids                                  0
 Number of duplicate encounters                               0
 Check for incorrect average visits                           0
 Check for the Proportion of no_shows where value is >1 or <0 0
```

Model Building

Establish train/test split and train control

*A custom summary function needs to be implemented here because the caret two class summary actually uses the first level for the event. This creates a mismatch with the beta coefficients which use the second level as the event during calculation. We could simply swap the reference level, but it is more intuitive to interpret the beta coefficients when churn is

used as the event, so this function is implemented to refactor the levels just for the summary. The end result is not tampered with which can be verified by viewing the confusion matrix.

```
library(caret)
```

```
Loading required package: lattice


Attaching package: 'caret'

The following object is masked from 'package:purrr':

    lift

The following objects are masked from 'package:fabletools':

    MAE, RMSE
```

```
set.seed(222)

train_indices = sample(1:nrow(churn_set), size = .7 * nrow(churn
train_data = churn_set[train_indices, ]
test_data = churn_set[-train_indices,]

fixed_summary <- function(data, lev = NULL, model = NULL) {

  # force both obs & pred to have same factor levels
  data$obs  <- factor(data$obs,  levels = c("Yes", "No"))
  data$pred <- factor(data$pred, levels = c("Yes", "No"))

  caret::twoClassSummary(data, lev = c("Yes", "No"), model = mod
}


train_control = trainControl(method = 'cv' , number = 10, classP
```

Train logistic regression model and observe results

```
model_cv = train(has_churn ~ prop_no_show + average_visits + cou
                 method = 'glm',
                 family = 'binomial',
                 trControl = train_control,
                 metric = 'ROC')
summary(model_cv)
```

```
Call:
NULL

Coefficients:
                                     Estimate Std. Error z
value Pr(>|z|)
(Intercept)                          1.946542   0.228119
8.533  < 2e-16 ***
prop_no_show                         2.913918   1.221871
2.385 0.017088 *
average_visits                      -0.071212   0.003567
-19.966  < 2e-16 ***
count_1                             -0.009451   0.002344
-4.032 5.54e-05 ***
count_2                             -0.031248   0.010281
-3.039 0.002371 **
count_3                              0.023931   0.007736
3.094 0.001977 **
count_4                             -0.005184   0.008692
-0.596 0.550874
count_5                              0.032412   0.042170
0.769 0.442132
EducationCollege                    -0.784029   0.234302
-3.346 0.000819 ***
`EducationHigh school diploma / GED` -0.749190   0.232979
-3.216 0.001301 **
`EducationLess than high school`     -0.805808   0.240879
-3.345 0.000822 ***
EducationUnknown                    -2.821062   0.612609
-4.605 4.12e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 5231.0  on 3806  degrees of freedom
Residual deviance: 4159.8  on 3795  degrees of freedom
AIC: 4183.8

Number of Fisher Scoring iterations: 5
```

view results of logistic regression model

```
model_cv$results
```

```
  parameter       ROC      Sens      Spec      ROCSD     SensSD
SpecSD
1      none 0.8083418 0.8600085 0.5835712 0.01891629 0.02454557
0.03926528
```

LDA model testing

- Marginally worse ROC and further drives down specificity.
- No reason to use as final model as it also has reduced explainability.

```
model_lda = train(has_churn ~ prop_no_show + average_visits + c
                  method = 'lda',
                  family = 'binomial',
                  trControl = train_control,
                  metric = 'ROC')
model_lda$results
```

```
  parameter       ROC      Sens      Spec     ROCSD     SensSD
SpecSD
1      none 0.8071513 0.8888581 0.5227497 0.0223665 0.02561104
0.04138112
```

Additional demographic testing

- For newly enrolled patients, all the clinic has to go on is demographic data. While our selected model focuses primarily on behavioral data, it is worthwhile to test a few extra demographics to see which could be significant.

Employment model

- Produced a similar ROC/Sensitivity/Specificity.
- Only significant differences in churn rates were for the unemployed and those not in the work force
- Unknown employment status was also significant, but that is more of a placeholder for NA values because we are not confident in setting a true default value for employment status.

```
model_employment =  train(has_churn ~ prop_no_show + average_vi
                          method = 'glm',
                          family = 'binomial',
                          trControl = train_control,
                          metric = 'ROC')

summary(model_employment)
```

```
Call:
NULL

Coefficients:
                              Estimate Std. Error z value
Pr(>|z|)
(Intercept)                   1.021905   0.062310  16.400  <
2e-16 ***
prop_no_show                  2.695339   1.182982   2.278
0.02270 *
average_visits               -0.067874   0.003546 -19.141  <
2e-16 ***
count_1                      -0.009568   0.002338  -4.092
4.28e-05 ***
count_2                      -0.030608   0.010176  -3.008
0.00263 **
count_3                       0.025079   0.007633   3.286
0.00102 **
count_4                      -0.006942   0.008726  -0.796
0.42627
count_5                       0.028537   0.042775   0.667
0.50468
`EmploymentNot in Workforce` -0.748383   0.268298  -2.789
0.00528 **
EmploymentRetired            -0.061341   0.214792  -0.286
0.77520
EmploymentSeasonal            0.148451   0.332235   0.447
0.65500
`EmploymentSelf Employed`     0.130985   0.162320   0.807
0.41969
EmploymentUnemployed          0.485392   0.086212   5.630
1.80e-08 ***
EmploymentUnknown            -3.079759   1.049679  -2.934
0.00335 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 5231.0  on 3806  degrees of freedom
Residual deviance: 4124.5  on 3793  degrees of freedom
AIC: 4152.5

Number of Fisher Scoring iterations: 5
```

```r
model_employment$results
```

```
  parameter       ROC      Sens      Spec      ROCSD     SensSD
SpecSD
1      none 0.8095632 0.8528905 0.5895092 0.02177453 0.01908476
0.04356681
```

Age model

- 18-19 year olds used as baseline
- Difference with older age ranges was barely significant and had a meager effect on churn chance

```r
model_age = train(has_churn ~ prop_no_show + average_visits + c
                  method = 'glm',
                  family = 'binomial',
                  trControl = train_control,
                  metric = 'ROC')
summary(model_age)
```

```
Call:
NULL

Coefficients:
                      Estimate Std. Error z value Pr(>|z|)
(Intercept)          -0.559969   0.808361  -0.693  0.48848
prop_no_show          2.829816   1.213680   2.332  0.01972 *
average_visits       -0.069316   0.003538 -19.594  < 2e-16 ***
count_1              -0.009687   0.002352  -4.118 3.82e-05 ***
count_2              -0.033091   0.010341  -3.200  0.00137 **
count_3               0.023137   0.007783   2.973  0.00295 **
count_4              -0.005857   0.008803  -0.665  0.50580
count_5               0.031559   0.042358   0.745  0.45625
`age_range20-29`      1.297930   0.814646   1.593  0.11111
`age_range30-39`      1.733475   0.811377   2.136  0.03264 *
`age_range40-49`      1.789054   0.811819   2.204  0.02754 *
`age_range50-59`      1.804631   0.812461   2.221  0.02634 *
`age_range60-64`      1.737265   0.815503   2.130  0.03315 *
`age_range65+`        1.947225   0.813196   2.395  0.01664 *
`age_rangeUnder 18`  13.126032 187.492603   0.070  0.94419
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)
```

```
    Null deviance: 5231.0  on 3806  degrees of freedom
Residual deviance: 4161.1  on 3792  degrees of freedom
AIC: 4191.1

Number of Fisher Scoring iterations: 11
```

```r
model_age$results
```

```
  parameter       ROC      Sens      Spec      ROCSD     SensSD
SpecSD
1      none 0.8086431 0.8557342 0.5906335 0.02699768 0.02418417
0.0428915
```

Sex Model

- No significant difference between men and women

```r
model_sex = train(has_churn ~ prop_no_show + average_visits + c
                  method = 'glm',
                  family = 'binomial',
                  trControl = train_control,
                  metric = 'ROC')
summary(model_sex)
```

```
Call:
NULL

Coefficients:
                 Estimate Std. Error z value Pr(>|z|)
(Intercept)      1.183160   0.059544  19.870  < 2e-16 ***
prop_no_show     2.856336   1.216336   2.348  0.01886 *
average_visits  -0.071271   0.003548 -20.088  < 2e-16 ***
count_1         -0.009688   0.002339  -4.142 3.44e-05 ***
count_2         -0.031915   0.010268  -3.108  0.00188 **
count_3          0.024748   0.007724   3.204  0.00135 **
count_4         -0.005025   0.008696  -0.578  0.56334
count_5          0.031286   0.042188   0.742  0.45835
GenderM          0.016955   0.076693   0.221  0.82503
GenderUnknown  -13.241155 221.154648  -0.060  0.95226
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 5231  on 3806  degrees of freedom
```

```
Residual deviance: 4186  on 3797  degrees of freedom
AIC: 4206

Number of Fisher Scoring iterations: 11
```

Results and Interpretations

- For our selected model, there were intuitive results such as the fact that the average number of encounters per year had a significant negative coefficient indicating that patients that engage more frequently are more likely to keep engaging. Likewise, there was a significant positive coefficient associated with the proportion of no-show encounters which confirms that patients that frequently miss appointments are more likely to churn. It should be noted that this effect is relatively small compared to other coefficients as the proportion can only fall between zero and one.
- Level of education had significant, albeit small differences between education levels. Patients with associates degrees who were used as the base line had the highest chance of churn. This is interesting because this level of education falls in the middle ground between total lack of education and a full college degree. The significance of the difference indicates that there could be a systemic effect driving the difference in churn rates, but it should be noted that the actual coefficients are low.
- Severity one and two diagnosis counts had negative coefficients which indicates a decreased chance of churn for patients that come in frequently with these conditions. However, the severity three diagnosis count had a significant positive coefficient which indicates that patients with these conditions would be more likely to churn. The cause of this requires more investigation. It is possible that patients are unsatisfied with their treatment or that the treatment necessitates unsatisfactory life style changes as in the case of diabetes. Another possibility is that the condition is inherently churn-driving to the instability it introduces into the lives of patients as in the case of depression and substance abuse.
- The results for severity four and five diagnoses were inconclusive because the features lacked significance.
- While the model did correctly identify the vast majority of churners (87%), there was a significant false positive rate with only 56% of non-churners being correctly identified. This is a limitation likely driven by new patients who have no behavioral data on record. The behavioral metrics for these patients like average number of

encounters per year and diagnosis counts will be zero, so the model will immediately identify them as churners despite them just enrolling. This is a limitation that will be further discussed in a later section.

Confusion matrix

```
predictions = predict(model_cv, test_data)
confusionMatrix(predictions, test_data$has_churn, positive = 'Ye
```

```
Confusion Matrix and Statistics

          Reference
Prediction  No Yes
       No  449 114
       Yes 305 764

               Accuracy : 0.7433
                 95% CI : (0.7213, 0.7643)
    No Information Rate : 0.538
    P-Value [Acc > NIR] : < 2.2e-16

                  Kappa : 0.4741

 Mcnemar's Test P-Value : < 2.2e-16

            Sensitivity : 0.8702
            Specificity : 0.5955
         Pos Pred Value : 0.7147
         Neg Pred Value : 0.7975
             Prevalence : 0.5380
         Detection Rate : 0.4681
   Detection Prevalence : 0.6550
      Balanced Accuracy : 0.7328

       'Positive' Class : Yes
```

Recommendations and limitations

- The current diagnosis groupings are a good start, but we do not have professional knowledge in medicine, so it would be best for [redacted] to work with their professionals to create better diagnosis groups. They can then monitor churn rates for each group to see which groups are

associated with an increased chance of churn. Diagnosis groups could be driving churn because patients are dissatisfied with treatment or because the conditions within the groups are inherently churn driving. In either case, [redacted] needs to be aware of which groups are driving churn to formulate strategies to mitigate their effects such as re-tuning treatments or simply expanding outreach to patients with these conditions. *Outside of the churn-driving conditions that we identified, engagement with the clinic was shown to drive further engagement. This demonstrates that it is critical for the clinic to be as active as possible in getting newly enrolled patients to schedule initial appointments. From our understanding, [redacted] already offers to schedule an appointment upon initial enrollment, but it would be best to send out additional reminders to patients who do not take up this offer.

- A key limitation of this model is that it relies heavily on behavioral data. It is not suitable for predicting churn for new patients without any behavioral data as mentioned prior. We did find that a patient's level of education had a significant but small effect on churn rates. In testing other models, we also found that employment and age range could also be explanatory, albeit only slightly. If [redacted] wants to determine a chance of churn for patients immediately upon enrollment, it would be best to develop a separate model based solely on these demographic features to determine which patients will be the most difficult in getting that initial engagement from. Once a patient has been enrolled long enough that their behavioral data is meaningful, their chance of churn can instead be predicted with a model like ours incorporating that behavioral data.

Naive Baseline Comparison

- Majority Class Baseline - This baseline output indicates as to what happens when the prediction for Churn is "Yes" for all the patients.
- The Naive prediction sits at a 53.8% Accuracy.
- Our model has (when used against unseen data) to achieve ~70% accuracy, making is a much better model compared to random predictions.

```
majority_class <- names(which.max(table(train_data$has_churn)))
naive_pred <- factor(rep(majority_class, nrow(test_data)), leve
confusionMatrix(naive_pred, test_data$has_churn, positive = "Ye
```

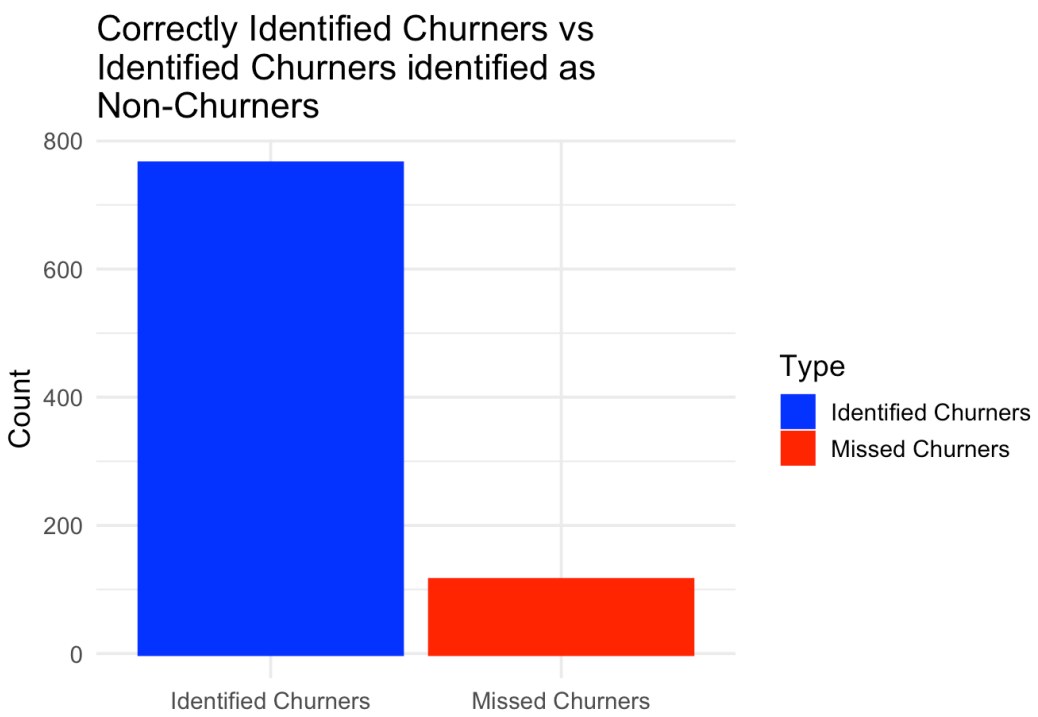
```
Confusion Matrix and Statistics
```

```
          Reference
Prediction  No Yes
       No    0   0
       Yes 754 878

               Accuracy : 0.538
                 95% CI : (0.5134, 0.5624)
    No Information Rate : 0.538
    P-Value [Acc > NIR] : 0.5102

                  Kappa : 0

 Mcnemar's Test P-Value : <2e-16

            Sensitivity : 1.000
            Specificity : 0.000
         Pos Pred Value : 0.538
         Neg Pred Value :   NaN
             Prevalence : 0.538
         Detection Rate : 0.538
   Detection Prevalence : 1.000
      Balanced Accuracy : 0.500

       'Positive' Class : Yes
```

Prediction visualizations

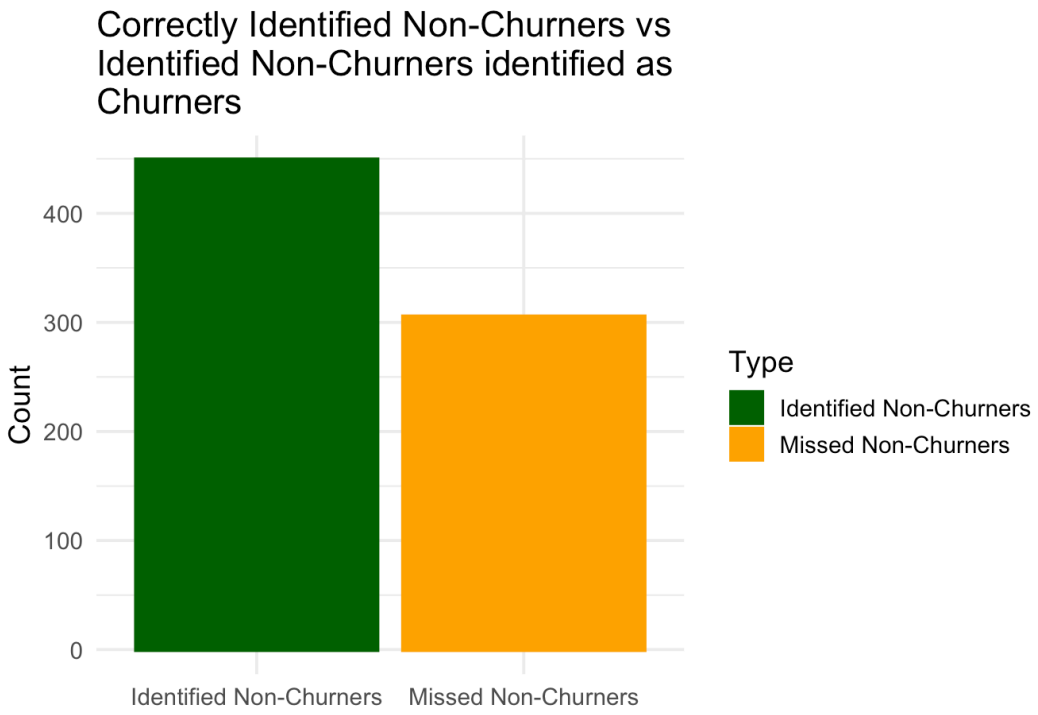
```
pred <- predict(model_cv, test_data)

cm <- confusionMatrix(pred, test_data$has_churn, positive = "Yes
tbl <- cm$table

TN <- tbl[1,1]
FP <- tbl[1,2]
FN <- tbl[2,1]
TP <- tbl[2,2]

df_tf <- data.frame(
  Type = c("Identified Churners", "Missed Churners"),
  Count = c(TP, FP)
)

df_nf <- data.frame(
  Type = c("Identified Non-Churners", "Missed Non-Churners"),
```

```
  Count = c(TN, FN)
)

colors_tf <- c("Identified Churners" = "blue",
               "Missed Churners" = "red")

colors_nf <- c("Identified Non-Churners" = "darkgreen",
               "Missed Non-Churners" = "orange")

ggplot(df_tf, aes(x = Type, y = Count, fill = Type, color = Type
  geom_bar(stat = "identity", linewidth = 1.1) +
  scale_fill_manual(values = colors_tf) +
  scale_color_manual(values = colors_tf) +
  labs(
    title = str_wrap("Correctly Identified Churners vs Identifie
    x = "",
    y = "Count"
  ) +
  theme_minimal(base_size = 14)
```




```
ggplot(df_nf, aes(x = Type, y = Count, fill = Type, color = Type
  geom_bar(stat = "identity", linewidth = 1.1) +
  scale_fill_manual(values = colors_nf) +
  scale_color_manual(values = colors_nf) +
  labs(
    title = str_wrap("Correctly Identified Non-Churners vs Iden
```

```
  x = "",
  y = "Count"
) +
theme_minimal(base_size = 14)
```



Actionable Insights Brief:

Executive Summary:

Patients at [redacted] Clinic with frequent engagement and low severity conditions were found to be the least likely to churn whereas patients that are being treated for moderate severity conditions such as diabetes, depression, and substance abuse were found to have increased chances of churn as they received treatment. [redacted] should work with their medical professionals on staff to refine diagnosis groupings and continue to monitor which diagnosis groups are statistically driving churn. For patients that lack diagnosis data and other behavioral data because they have enrolled but not yet engaged, [redacted] can use demographic features such as the level of education to get a sense for which patients need to be pursued most actively.

Problem Definition:

Patient churn is best defined as when an enrolled patient disengages with [redacted] Clinic whether officially or unofficially. This is pertinent to a non-profit clinic because the objective is to provide medical care which is not possible if a patient disengages. We argue the best definition for churn is when a patient goes longer than a year without a completed encounter or when a patient is disenrolled or allows their enrollment to expire without valid reason. Valid reasons include going on Medicare or Medicaid as well as private insurance. We exclude the reasons from our churn definition because in these cases the patient has exited [redacted] area of service, but they are still receiving medical care from other sources.

Methodology:

To model patient churn, we used logistic regression. We used the average number of visits per year and the proportional no show rate as well as patient level of education as explanatory features. We also wanted to observe how churn rates differ depending on the severity of the conditions a patient is being treated for. To do this, we grouped diagnoses into five severity categories and then used the counts of these categories for each patient as a feature. Severity one diagnoses include mild uncategorizable conditions such as dry eye and mild pain. Severity two diagnoses include more recognizable albeit less threatening conditions such as chronic pain and pre-diabetes. Severity three includes more serious but not necessarily life-threatening conditions such as diabetes, depression, and substance abuse. Severity four includes more threatening conditions such as heart disease. Severity five includes exclusively cerebrovascular conditions.

Key Findings:

Logistic regression assigns weights to each feature as well as a value that indicates the likelihood that the weight is non-existent referred to as significance. A full list of the features used in the model as well as their weights and whether the weight is likely to be true is provided in the table below:

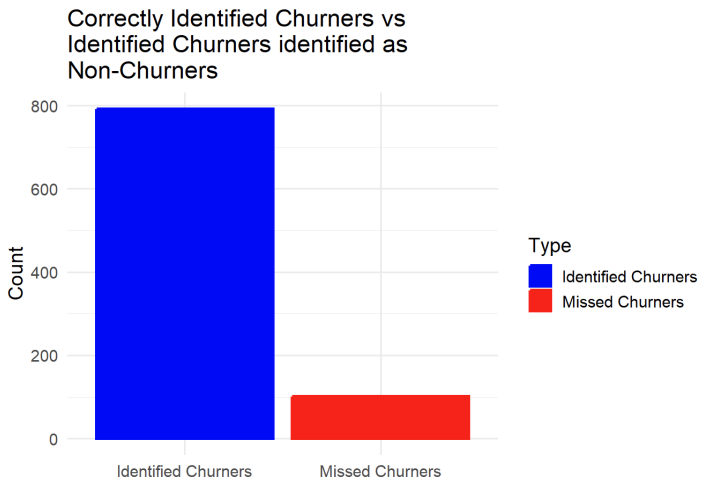
Feature	Weight	Significance
Average Encounters Per Year	**-0.067**	**Yes**
Proportion of No-Shows	**2.960**	**Yes**
Severity One Count	**-0.010**	**Yes**
Severity Two Count	**-0.029**	**Yes**
Severity Three Count	**0.023**	**Yes**
Severity Four Count	-0.007	No

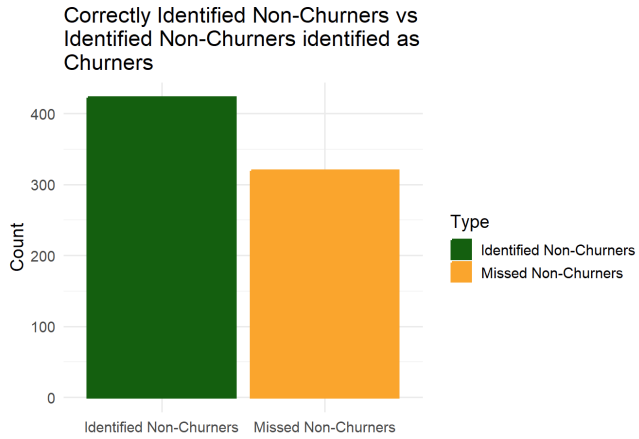
Severity Five Count	0.029	No
Less Than High School	**-0.880**	**Yes**
High School	**-0.778**	**Yes**
College	**-0.853**	**Yes**

Some of the weights assigned to features are intuitive such as for average encounters per year and proportion of no-show encounters. It makes sense that a patient that comes to the clinic more often is more likely to keep coming back and that a patient who continually misses appointments without reason is likely to stop engaging with the clinic soon. Similar insights can be gleaned from the counts for severity one and two diagnoses. The negative weight indicates that patients that come in more frequently with low severity conditions are less likely to churn. What is troubling is the significant positive weight assigned to severity three conditions. It indicates that as a patient receives treatment for these more severe conditions, they are more likely to stop seeking treatment. There are a few possible explanations for this. One is that patients are simply dissatisfied with treatments for these conditions and are therefore looking elsewhere for treatment or stop seeking treatment altogether. Another possibility is that conditions such as depression and substance abuse destabilize the lives of patients to the point where they stop seeking treatment.

The weights assigned to education levels were also interesting. Associate degrees, which were used as the baseline hence why they are not shown in the table above, were associated with the highest chance of churn by a significant margin. This may be because of the nature of the work that people with associates degrees are typically in or perhaps the level of income that people with associates degrees typically

have. We do not have enough information to ascertain the true cause, but it is worth noting that the clinic may need to be more active in retaining patients with an associates. The model results in testing can be viewed in the graphs below:





When tested, the model had an accuracy of 74%. 87% of churners within the data set were correctly identified as churned and 56% of non-churners were correctly identified as not churned. The capture rate of churners is promising but the false positive rate is a limitation that will be discussed later in the brief.

Recommendations and Limitations:

We believe that the features with the most need for further investigation are the diagnosis counts by group. This analysis grouped diagnoses by severity, but these groupings could be improved by medical professionals. Once refined, these groupings can be used to monitor which groups of conditions are driving churn. In this case, we found that severity three diagnoses were driving churn which indicates that [redacted] may

need a strategy to either improve patient satisfaction towards their treatments or to mitigate the effect of conditions that may be inherently churn-driving such as depression.

A critical limitation of this model is that it relies primarily on patient behavioral data to make predictions. This means that predictions will be far less reliable for patients that are freshly enrolled with the clinic because they have no behavioral data associated with them. We recommend that the clinic adopt a model to estimate chance of churning for new patients using exclusively demographic features. This would provide an initial estimate of how active [redacted] will need to be in getting initial engagement from a freshly enrolled patient. Once the patient has been enrolled for some time, they can be transitioned over to the behavior-based model which should provide a more detailed estimate of churn.

In conclusion, we recommend that [redacted] Clinic further monitor and address the effect of diagnoses on churn rates, particularly diagnoses for mildly severe conditions such as depression and substance abuse, and that they adopt two churn prediction models, one such as ours to estimate the chance of churn based on patient behavior and engagement and one based purely on demographic data to estimate the chance of churn at the time of enrollment.